MD AL EMRAN HOSSAIN

Level-16, Tower-04, Suvastu Nazar Valley (Apartment), Badda-1212, Dhaka
+880 1776-154462 | emrandocs@gmail.com
LinkedIn | Google Scholar | GitHub | Portfolio

CAREER OBJECTIVE

Senior Machine Learning Engineer and AI/ML Team Lead with **4.5+ years** of industry experience building and deploying production-grade AI systems for FinTech, banking, and enterprise automation. I specialize in **Generative AI, Document AI, Multimodal AI, NLP, Computer Vision, OCR, RAG, and Agentic AI,** with experience taking solutions from business requirements and POCs to architecture, deployment, and production. Currently **leading a 6+ member ML team** and delivering secure, resource-efficient AI solutions across SaaS and restricted on-premises environments. My work combines practical enterprise AI engineering with continuous research in multimodal learning, reinforcement learning, document intelligence, and intelligent automation.

EDUCATION

- **BRAC University,** Bangladesh — Jan 2026 – Present
  MSc in Computer Science and Engineering
  Running Courses: Generative Machine Learning, Advanced Parallel Algorithms
  Completed Courses: Advanced AI, Advanced NLP, Advanced RL, Computer Vision, and Big Data Analytics
- **United International University (UIU),** Bangladesh — Feb 2018 – Mar 2022
  BSc in Computer Science and Engineering — **CGPA : 3.90 / 4.00**
  Scholastic Award: **Summa Cum Laude**
  Courses: Artificial Intelligence, Machine Learning, Pattern Recognition, Bioinformatics, Probability & Statistics

PROFESSIONAL EXPERIENCE

Jan 2025 – Present | Senior ML Engineer & ML Team Lead
Spectrum Software & Consulting (Pvt.) Ltd.

- Lead a 6+ member ML team and drive AI/ML initiatives across banking, FinTech, RMG, and enterprise automation.
- Engage with 10+ banks and 5+ financial organizations to identify AI/ML opportunities and deliver POCs and production solutions.
- Architect and deploy low-resource AI solutions using LLMs, VLMs, OCR, RAG, and Document AI across SaaS and secure on-premises environments.
- Lead AI product strategy, technical proposals, tenders, solution costing, and pricing for products including e-KYC, XtractAI, and SpectraTrade.
- Oversee model development, fine-tuning, MLOps, deployment, and production integration.
- Introduced internal AI/ML productivity tools, including Semi-Auto Annotator and QA Automation.

Jul 2022 – Dec 2024 | Machine Learning Engineer
Spectrum Software & Consulting (Pvt.) Ltd.

- Developed and deployed production AI/ML solutions for banking, FinTech, trade finance, and manufacturing.
- Built Document AI pipelines for classification, OCR, information extraction, and table/document understanding.
- Fine-tuned models including LayoutLMv2, Donut, YOLO, and LLaMA for enterprise financial use cases.
- Developed ML APIs and integrated AI solutions with enterprise applications and RPA workflows.
- Built domain-specific datasets, annotation pipelines, and human-in-the-loop feedback systems.
- Contributed to model evaluation, monitoring, retraining, and secure on-premises deployment.

Feb 2022 – Jun 2022 | Research Assistant
United International University (UIU)

- Conducted research on cyberbullying detection across social media platforms under the supervision of Dr. Saddam Hossain Mukta.
- Contributed to a funded project by the Institute for Advanced Research (IAR), involving data collection, preprocessing, and model experimentation.

Nov 2021 – Feb 2022 | Undergraduate Teaching Assistant
United International University (UIU)

- Supported *Pattern Recognition* and *Database Management* labs by guiding student projects and assisting with coursework.
- Evaluated assignments, quizzes, and lab reports, ensuring timely feedback and academic support.

RESEARCH & PUBLICATIONS

- **Undergraduate Thesis** [2024]
  Deep Transfer Learning Based Peer Review Aggregation and Meta-Review Generation for Scientific Articles
  arXiv preprint arXiv:2410.04202
  DOI: 10.48550/arXiv.2410.04202
- **Journal Article** [2023]
  A Review on Deep-Learning-Based Cyberbullying Detection
  Future Internet, 15(5), 179, 2023
  DOI: 10.3390/fi15050179
- **Conference Paper** [2022]
  Adaptive Tabu Dropout for Regularization of Deep Neural Networks
  International Conference on Neural Information Processing (ICONIP), pp. 355–366, Springer, 2022
  DOI: 10.1007/978-3-031-30105-6_30

PROJECTS

Industry Projects & Experimental Work

Agentic KYC Platform (Proof of Concept) [2026-Present]
End-to-End AI/ML Developer

- Designed and built an interactive end-to-end prototype of an agentic AI system that automates periodic KYC (Know-Your-Customer) re-verification for a Bangladeshi commercial bank, based on a client-signed BRD.
- Modelled a human-in-the-loop architecture: the AI agent selects due customers, runs notification and reminder campaigns, reads handwritten Bangla/English forms (OCR with confidence thresholds), compares previous vs new KYC and runs logical-consistency checks. Every core-banking (T24) change still needs approval from a named Operations officer.
- Built rule-based consistency checks (limit vs income, profession-change plausibility, source of funds vs profession, required documents) that send clean cases to straight-through processing and raise flagged cases for human review.
- Cut human interaction in the demo flow from 19 clicks to 4 decision points (Approve, Request info, Block, Escalate) and added a "human touches per 100 customers" KPI.

XtractAI – SSCL's Document AI Platform ***(ML Lead)*** [2025-Present]
(Ongoing development: continuously integrating new Document AI capabilities)

- Built a live demo platform to showcase SSCL's in-house capabilities in Document AI, enabling real-time document processing and intelligent information extraction.
- Led the development and deployment of the machine learning components, including model integration and backend processing pipelines.
- Enabled automated extraction of key data from diverse document types (e.g., PDFs, forms, images), demonstrating practical AI solutions for enterprise use cases.
- Provided live product access for evaluation with *1,000 free credits* upon Gmail login for testing and validation purposes.

e-KYC – Electronic Know Your Customer System (Deployed in *TAP App*) [2025-2026]
[System Architect | Engineering Manager | Lead Developer]

- Led end-to-end product strategy, architecture, and development of a production e-KYC platform, currently live across 3 banks and 2 financial organizations for secure digital onboarding.
- Integrated with the Bangladesh Election Commission for NID verification and facial matching.
- Built an advanced image-processing + OCR pipeline with dewarping, blur/light detection, angle correction, enhancement, and optimized GPU inference, reducing OCR latency from 16 seconds to <1 second.
- Fine-tuned NID-specific models achieving 97.48%+ English and 91.23% Bangla extraction accuracy using Exact Matching on real banking data.
- Implemented video liveness detection, replay-attack prevention, and live-to-NID face matching for identity-fraud prevention.
- Designed an administrative portal with analytics, monitoring, and role-based access control.

Import Document Scrutiny *(Project Lead | System Architect)* 2025
(*Achieved 92% reduction in cost and manual effort)*

- Automated cross-validation of MT700 Letter of Credit (LC) clauses against submitted import documents.
- Validated the structure and completeness of import documents in accordance with UCP 600 and ISBP 821 rulebooks using a RAG-based framework.
- Converted unstructured LC messages into executable business logic and verified them against automatically extracted data from classified trade documents.
- Implemented a multi-modal AI approach using a fine-tuned GPT-OSS-20B model integrated with DotOCR for accurate text and layout understanding.

SpectraTrade – Financial Document Classification & Extraction ***(ML Lead)*** [2023-2025]

- Achieved high-accuracy financial document classification and information extraction using GPT, LLaMA (3.5 Vision finetuned), Qwen, Gemma, and Donut models.
- Optimized processing speed to 3–5 seconds per page, improving efficiency for large-scale banking operations.
- Deployed in ***PrimeBanijjo***, the country's first AI-powered digital LC management platform, enabling faster, more accurate, and seamless trade operations.

DockStream – RMG Sector Automation ***(Project Manager & ML Lead)*** [2024-2025]

- Led automation of document classification, data entry, form filling, and information submission across multiple websites using RPA technologies.
- Designed adaptive pipelines for extracting data from structured and unstructured documents, enabling end-to-end process automation.
- Achieved an 86% reduction in manual effort, human error, and associated operational costs.

SQA Automation [2025]
- Developed an internal productivity tool for QA operations, adapting to dynamic UI and operational changes based on client requirements.
- Minimized manual test case writing after each change, significantly reducing QA effort and accelerating release cycles.
- Developed as an internal productivity tool.

Robotic Process Automation (RPA) & GenAI for Financial Document Processing [2023-2025]
- Led OCR pipeline development supporting **24+ financial document** types with multi-format JSON outputs.
- Extracted key data from financial documents using Faster R-CNN, Donut, YOLOv5/v8, LayoutLMv2, and fine-tuned Donut and LLaMA models for improved accuracy.
- Developed financial document classification and table extraction (detection & segmentation) leveraging LayoutLMv2 and related techniques.

Semi-Auto Annotator & DocTrainer [2025]
- Developed a tool for automated PDF annotation with human-in-the-loop feedback (RLHF).
- Enabled iterative model retraining for continuous accuracy improvement.

MLOps Experiences

eKYC Platform – SaaS-Based Identity Verification Solution [2026]
- Led end-to-end MLOps deployment of a SaaS-based eKYC platform, deploying NID OCR, Liveness Detection, and EC Verification services through secure APIs with scalable inference infrastructure, load balancing, and NGINX-based hosting.

SpectraTrade – Secured Banking Deployment *(Leading Private Bank in Bangladesh)* [2023-2024]
- Led end-to-end MLOps deployment of ML modules (LC Management, Shipping Guarantee) on RHEL-secured on-premises infrastructure.
- Established the initial ML runtime environment, managed model lifecycle across UAT and Production, and integrated ML services via secure, API-based interfaces.

DockStream - RMG Sector Automation (Enterprise ML Deployment) [2024–2025]
- Led end-to-end MLOps deployment of ML-driven document intelligence and RPA automation pipelines for large-scale RMG operations.
- Operationalized document classification, structured/unstructured data extraction, and form automation models, integrating ML services with RPA workflows via production-grade APIs, resulting in an 86% reduction in manual effort and operational errors.

Academic Research & Projects

ZeroDepthFAS – Passive Human Face Liveness Detection [MSc Course Work & Ongoing Research] [2026]
- Built a passive face anti-spoofing system fusing Vision Transformer (Large) with EfficientNet-B0 for real-time e-KYC liveness detection.
- Achieved AUC 1.000 in-distribution and HTER 1.48% on zero-shot out-of-distribution replay attacks; deployed as a REST API with sub-second GPU inference.

HealthCareGPT – LLM + RAG for Disease Prediction [Course Work] [2024]
- Designed a healthcare assistant for disease prediction & personalized recommendations using LLMs and Retrieval-Augmented Generation (RAG).
- Demonstrated higher accuracy than traditional ML methods.

RLHF-Enhanced Document Classification (LayoutLM) [Course Work] [2024]
- Integrated Reinforcement Learning with Human Feedback (RLHF) to improve LayoutLM-based financial document classification.

UIU EMAS – Event Management & Archiving System [Team Project] [2022]
- Web-based system for managing the CSE Project Show & Club Forum Fest.
- Designed as part of a Software Engineering course project.

TECHNICAL EXPERTISE

- **Programming Languages:** Python*, C/C++, Java, PHP, Assembly
- **Machine Learning & Deep Learning**
  - **Libraries & Frameworks:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face (Transformers, Datasets, Accelerate)
  - **ML & DL Experience:** Linear/Logistic Regression, Decision Trees, Random Forests, SVM, KNN, K-Means, Naive Bayes, PCA, XGBoost, CNN, RNN, LSTM, Transformers, RL, RLHF, BERT, Donut, YoLo, etc.
- **Generative AI Ecosystem**
  - **Fine-Tuning LLMs/VLMs:** LLaMA, Qwen, Mistral, Granite, DeepSeek, Gemma, GPT-OSS
  - **API-Based LLMs:** OpenAI, Anthropic (Claude), Google Gemini, Alibaba Dashcope, Together AI
  - **OCR Tools:** DeepSeek-OCR, Chandra, Surya, LightOn, Paddle, EasyOCR, Tesseract, TrOCR, olmOCR, etc.
  - **Frameworks & Tools:** LangChain, Hugging Face Transformers, Prompt Engineering, Vision Transformers, MCP, MLOps Workflows
  - **RAG & Knowledge Systems:** RAG Pipelines (including Graph RAG with Neo4j), Vector Databases (Pinecone, FAISS), Adaptive Chunking Strategies

- **Deployment, Inference & Application Frameworks**
  - **Inference Engines:** vLLM, Ollama, Together AI, TensorRT
  - **Web & API Frameworks:** Flask, FastAPI, Gradio, Django
  - **Cloud:** AWS (foundational experience), S3, API Development & Hosting
  - **Workflow Automation & Orchestration:** n8n, CI/CD Workflows, Jira
- **Testing, Scripting & Agents:** Playwright, Selenium, PyTest, Computer-Use Agents (CUAs)
- **Databases & Storage:** MySQL, PostgreSQL, SQLite
- **DevOps & Version Control:** Git, GitHub, GitLab, Docker, Nginx
- **Operating Systems:** Linux (Ubuntu, RHEL/CentOS), Windows
- **GPU Setup & Used:** Nvidia A4000, A4500 Ada, A5000, A6000, DGX Spark (ARM64)
- **Data Engineering & API Development:** Data Preprocessing, Feature Engineering, API Development, Integration of AI models into production pipelines

ACHIEVEMENTS

- Scholastic Award **Summa Cum Laude** at 7th Convocation of UIU. [2024]
- Awarded a CGPA-based tuition fee waiver for the MSc program at BRAC University. [2026-Present]
- Out of 12 trimesters, 8x100% and 3x50% Scholarship for outstanding results at UIU. [2018-2022]
- **Champion** in UIU CSE Project Show (Fall 2022) for "*UIU EMAS*". [2022]
- **2nd Runner-Up** in UIU CSE Project Show for "*Pick-a-Book*". [2021]
- **General Board Scholarship** in *HSC* exams. [2016]
- **Talent Pool Scholarship** (9th district rank) in *SSC*. [2014]

EXTRACURRICULAR

- **Instructor** at Programmer Doctor's Chamber, addressing programming challenges. [2021-2022]
- **Programming Wing** Executive and **Media & Communication** Executive at *UIU Computer Club*. [2018-2022]
- **Founding Member** of *UIU Mars Rover Team*. [2021]
- **General Member** of *UIU Robotics Club*. [2018-2022]
- **Volunteer** at *ICAICT 2020* and various national/international events. [2022]
- Former Member of *Bangladesh Biggan Porishod*. [2016-2018]
- Former **Scout Group Leader** at *Shahebrampur High School*. [2012-2014]